Enterprise Process Automation and Software Quality Assurance Frameworks in BHEL's Digital Infrastructure

A REPORT

submitted by

Shreya Maitra(23BRS1283)

in partial fulfilment for the award

of

B. Tech. Computer Science and Engineering (Artificial Intelligence & Robotics)

School of Computer Science and Engineering



VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

October - 2026



School of Computer Science and Engineering

DECLARATION

I hereby declare that the project entitled **"Enterprise Process Automation and Software Quality Assurance Frameworks in BHEL's Digital Infrastructure"** submitted by me to the School of Computer Science and Engineering, Vellore Institute of Technology, Chennai Campus, Chennai 600127 in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology – Computer Science and Engineering (Artificial Intelligence & Robotics)** is a record of bonafide work carried out by me**.** I further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma of this institute or of any other institute or university.

Signature

Shreya Maitra (23BRS1283)



School of Computer Science and Engineering

CERTIFICATE

The project report entitled "**Enterprise Process Automation and Software Quality Assurance Frameworks in BHEL's Digital Infrastructure**" is prepared and submitted by **Shreya Maitra (Register No: 23BRS1283)**. It has been found satisfactory in terms of scope, quality and presentation as partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology – Computer Science and Engineering (Artificial Intelligence & Robotics)** in Vellore Institute of Technology, Chennai, India.

Examined by:

Examiner I **Examiner II**

INSERT THE CERTIFICATE OF MERIT OBTAINED FROM THE INDUSTRY WHERE YOU DID YOUR INTERNSHIP

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of this project would be incomplete without acknowledging the consistent cooperation, valuable guidance, and encouragement of the individuals who helped crown all efforts with success.

First and foremost, I express my sincere gratitude to **Dr. D. Sathian**, Head of the Department (HoD), B.Tech. Computer Science and Engineering (Artificial Intelligence & Robotics), Vellore Institute of Technology, Chennai, for his continuous support and guidance throughout the period of this training.

I extend my heartfelt thanks to **Dr. Viswanathan V**, Dean, School of Computer Science & Engineering, Vellore Institute of Technology, Chennai, for providing the institutional support and environment conducive to learning and technical development.

I am deeply thankful to **Dr. Nithyanandam P**, Associate Dean, School of Computer Science & Engineering, Vellore Institute of Technology, Chennai, for his constant encouragement and valuable insights.

My sincere thanks also go to **Dr. Suganya G**, Associate Dean, School of Computer Science & Engineering, Vellore Institute of Technology, Chennai, for her support and motivation during this process.

I would like to express my appreciation to **Dr. Sweetlin Hemalatha C**, Associate Dean, School of Computer Science & Engineering, Vellore Institute of Technology, Chennai, for her continuous encouragement throughout my academic endeavors.

I am deeply indebted and thankful to **Mr. Ram Ashish Singh Kushwaha** (Sr. Manager / Digital Transformation) and all the concerned officers and staff of **Bharat Heavy Electricals Limited, HERP Plant, Varanasi (Uttar Pradesh)** for their valuable guidance, constant advice, and mentorship, without which I could not have successfully completed my summer training.

Last but not least, I express my heartfelt gratitude to the workers of Bharat Heavy Electricals Limited who always gave their full cooperation in helping me understand the multiple production operations and practical workflows in the plant. I am also thankful to my parents and friends for their constant moral support and encouragement.

CONTENTS

Chapter	Title	Page
	Title Page	i
	Declaration	ii
	Certificate	iii
	Industry certificate	iv
	Acknowledgement	v
	Table of contents	vi
	List of Tables (insert only if applicable)	vii
	List of Figures (insert only if applicable)	viii
	List of Abbreviation	ix
	Abstract	x
1	Introduction	01
2	Chapter Name	02
3	Chapter Name	03
4	Conclusion	04
	References	05
	Appendix – I (Sample source code, screenshots, etc)	06
	Any other Appendix (pictures of your model, photos of the model, etc)	
	…….	

ABSTRACT

Bharat Heavy Electricals Limited (BHEL) is India's largest government-owned power generation equipment manufacturer, and its Heavy Equipment Repair Plant (HERP) at Varanasi supplies O&M spares, repairs turbine components and manufactures load hangers for power stations across the country. Like many large public sector manufacturing units, HERP is moving towards digital enablement, with its Digital Transformation Group modernizing how shop floor data is captured, stored and reported. At present, however, much of the operational data, including machine usage schedules, inventory depletion, spare part allocation, maintenance timestamps and shift-wise production targets, still resides in standalone Excel workbooks, departmental databases and physical registers.

This fragmented setup has clear limitations. Retrieving a record, such as checking spare part availability against an active work order, requires opening multiple files and scanning rows manually, which is effectively an O(n) linear search. Duplicate part code entries from dual data entry, inconsistent date formats and null quantity fields reduce reliability, while manual reconciliation introduces errors and delays between shop floor activity and management reporting. In addition, any software introduced into BHEL's internal network must meet strict security and self-hosting requirements, and there was no structured basis for choosing testing tools to validate new scripts before deployment.

The problem addressed in this work is therefore twofold: reducing data retrieval latency and manual processing errors in HERP's production logs, and establishing a dependable software quality assurance approach so that automated solutions can be deployed safely in a live industrial environment.

The proposed solution has three components. First, a Python and Pandas based ETL pipeline performs deduplication, null handling, type casting and key normalization,

converting fragmented spreadsheets into a unified, hash-indexed dataset keyed on part code. This shifts lookups from O(n) to O(1) at the cost of modest additional memory, roughly 5–10 MB for 10,000 records. Second, a comparative evaluation of Selenium WebDriver, PyTest and Jenkins, assessed on functional coverage, data processing capability, security posture, integration compatibility and maintenance overhead, led to a layered QA architecture. In it, a self-hosted Jenkins pipeline triggers PyTest for backend logic and Selenium for dashboard validation on every code commit, and deployment is approved only if all tests pass. Third, an automated decision support architecture channels the cleaned data into a central repository and a BI dashboard that provides KPI monitoring, exception alerts and trend reports. Together, these measures reduce process cycle time, improve data integrity and give plant management timely, reliable information for decision-making.

1 INTRODUCTION

Introduction to BHEL

Bharat Heavy Electricals Limited (BHEL) is an Indian central public sector
undertaking and the **largest government-owned power generation equipment manufacturer**. It is under the ownership of the Government of India and administrative control of the Ministry of Heavy Industries. Established in **1964**, BHEL is based in New Delhi with the help of Soviet technology.

It has a comprehensive portfolio of products, systems and services in the areas of power-thermal, hydro, gas, nuclear & solar PV; transmission; transportation; defence & aerospace; oil & gas and new areas like BESS and EV chargers.

The worldwide installed base of power generating equipment supplied by BHEL **exceeds 194 GW**, making it the undisputed leader amongst Indian power plant equipment manufacturers. Having installed **more than 1000** thermal, hydro, nuclear, gas and solar PV based power generating sets in the country, the company is actively engaged in building strong foundations for cleaner and greener energy generation and usage in future with critical technologies coal to methanol.

Establishment of BHEL in 1964 was a breakthrough for upsurge in India's Heavy Electrical Equipment industry. Consistent performance in a highly competitive environment enabled BHEL to attain the **coveted 'Maharatna' status in 2013.**

BHEL as a part of Pt. Jawaharlal Nehru's vision was bestowed with the onus to make the country self-reliant in manufacturing of heavy electrical equipment.

This dream has been more than realized and the contribution in nation building endeavors is going to continue likewise.

CHAPTER 4:

Enterprise Process Automation and Software Quality Assurance Frameworks in BHEL's Digital Infrastructure

4.1 Introduction to Digital Enablement at BHEL HERP

Bharat Heavy Electricals Limited (BHEL) is one of India's largest and oldest public sector engineering enterprises, established in 1964 under the Ministry of Heavy Industries. With a presence across more than 17 manufacturing divisions and 180+ project sites nationwide, BHEL forms the backbone of India's capital goods sector — contributing to power generation, transmission, transportation, and industrial systems. The Heavy Equipment Repair Plant (HERP) at Varanasi is one of BHEL's most strategically critical manufacturing facilities, responsible for maintaining, overhauling, and repairing high-precision mechanical sub-assemblies used in turbines, generators, heat exchangers, and pressure vessels deployed at power plants across the country.

At a facility of this scale, the volume and complexity of manufacturing data is immense. Shop floor operations run across multiple shifts, involving hundreds of machine tools, thousands of component part numbers, and intricate interdependencies between procurement, inventory, fabrication, quality inspection, and dispatch. Coordinating these workflows without a robust digital infrastructure leads to scheduling conflicts, documentation backlogs, and costly production delays.

To address this, BHEL HERP operates a dedicated Digital Transformation (DX) Group — an internal engineering division tasked with modernizing the plant's data handling capabilities. The DX Group designs and deploys automated data pipeline models,

analytics dashboards, and software quality assurance (QA) frameworks that digitize the plant's core operational workflows. The group's mandate spans three primary objectives:

1. Staging unit-level manufacturing data safely and systematically into centralized repositories
2. Minimizing latency between plant floor operations and management-level reporting
3. Eliminating manual data processing exceptions that introduce errors and slow down decision-making

This chapter details the specific technical initiatives undertaken by the DX Group during the internship period, covering algorithmic optimization of production logs, evaluation of open-source QA tools, and the design of an automated decision support architecture.

4.2 Algorithmic Optimization of Production Logs

4.2.1 The Data Engineering Challenge

The manufacturing floor at HERP generates a continuous stream of operational data: machine tool usage schedules, real-time inventory depletion records, spare parts allocation logs, maintenance event timestamps, and shift-wise production targets. Historically, this data was stored across fragmented legacy modules — a combination of standalone Excel workbooks, departmental databases, and physical registers — which made cross-referencing information across systems slow and error-prone.

The core problem was one of data retrieval latency. When a production supervisor needed to check spare part availability against an active work order, the process involved manually opening multiple spreadsheets, cross-referencing part codes by scrolling

through row-by-row lists, and reconciling discrepancies by hand. This is a textbook O(n) linear search operation — where the time taken to retrieve a data point scales proportionally with the total number of records in the dataset.

4.2.2 Algorithmic Solution: Linear to Relational Indexing

The DX Group addressed this inefficiency through Python scripting to transform fragmented spreadsheet data into structured, indexed datasets. By converting raw tabular dumps into relational key-value structures — essentially building lookup dictionaries and indexed dataframes using libraries like Pandas — retrieval operations were shifted from linear scanning to constant-time lookups:

Search Latency Improvement:

$O(n) \rightarrow O(1)$

Where:

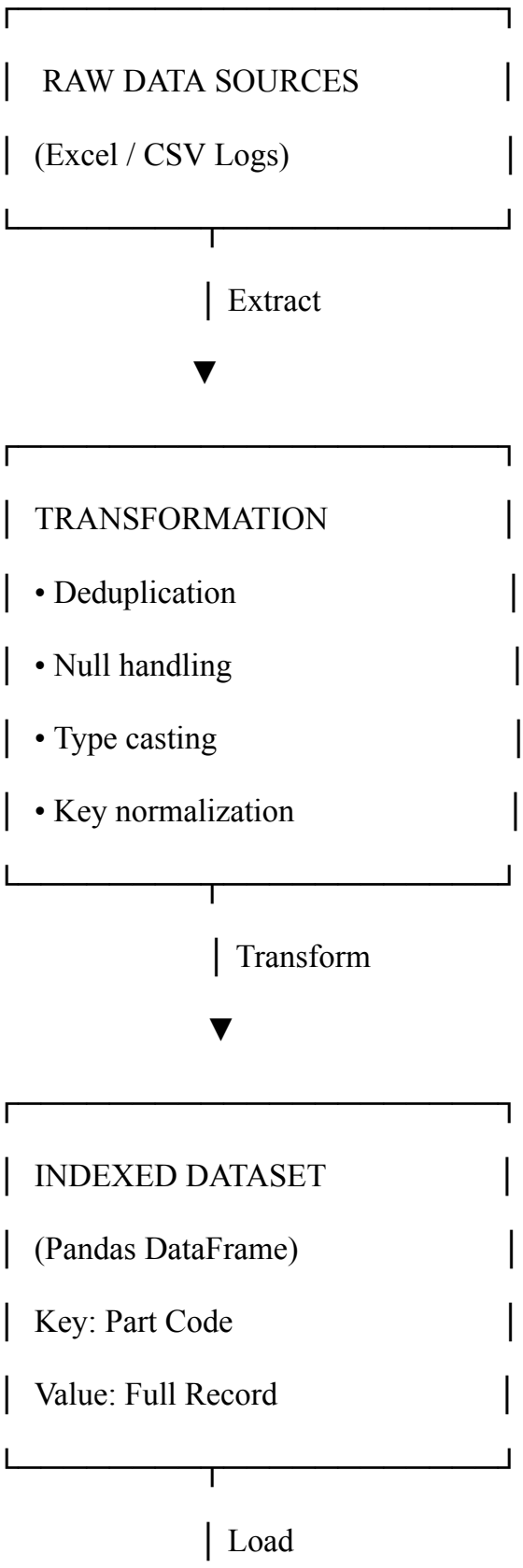
- O(n) = linear scan across n records (legacy method)
- O(1) = direct key-based lookup via indexed dictionary or hash table (optimized method)

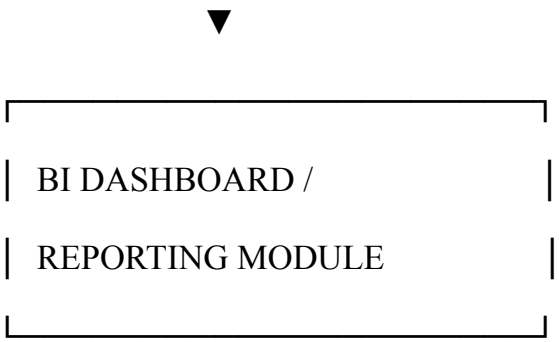
For a dataset containing 10,000 inventory records, a linear search might evaluate up to 10,000 comparisons per query. An indexed lookup evaluates exactly one. At the scale of HERP's operations — where dozens of such queries are run per shift — this improvement translates into measurable reductions in process cycle time.

4.2.3 Implementation: Unified Dataset Architecture

The Python scripts developed during this internship followed a structured ETL (Extract, Transform, Load) pipeline:

FIGURE 1: ETL Pipeline — Production Log Optimization


RAW DATA SOURCES
(Excel / CSV Logs)
Extract
TRANSFORMATION
• Deduplication
• Null handling
• Type casting
• Key normalization
Transform
INDEXED DATASET
(Pandas DataFrame)
Key: Part Code
Value: Full Record
Load



The transformation stage handled common data quality issues found in legacy HERP logs: duplicate part code entries caused by dual data entry across departments, inconsistent date formats across different shop floor registers, and null values in quantity fields due to incomplete shift logging. Once cleaned and indexed, the unified dataset served as the single source of truth for all downstream reporting.

4.2.4 Space Complexity Trade-offs

The shift from sequential scanning to hash-indexed lookup does involve a trade-off: O(n) additional memory is required to maintain the index structure in RAM. At HERP's data volumes, this is acceptable — a 10,000-row inventory dataset with 15 columns occupies approximately 5–10 MB in memory, well within the capacity of standard enterprise workstations. The time savings gained from eliminating sequential search far outweigh the marginal memory overhead.

TABLE 1: Algorithmic Complexity Comparison

Operation	Legacy Method	Optimized Method	Improvement
Single record lookup	O(n) — linear scan	O(1) — hash lookup	~10,000x faster for 10k records
Multi-rec	O(n²) —	O(n) —	Significant

ord aggregation	nested loops	vectorized Pandas	for large datasets
Dataset join (two sources)	O(n × m) — manual cross-ref	O(n + m) — merge on key	Near-linear instead of quadratic
Space consumption	O(1) — no index stored	O(n) — index in memory	Minor trade-off, acceptable

4.3 Evaluation Framework for Open-Source Testing Utilities

4.3.1 Why QA Frameworks Matter in Industrial IT

Deploying software within BHEL's internal enterprise network carries significantly higher stakes than typical commercial software rollouts. Errors in production logging scripts or dashboard queries can propagate incorrect data into management systems, triggering faulty procurement decisions or missed maintenance schedules. For this reason, the DX Group follows a rigorous software validation process before any new script or application is integrated into live systems.

During the internship, a systematic evaluation was conducted of three widely-used open-source QA and automation frameworks to assess their compatibility with HERP's security constraints, infrastructure topology, and testing requirements.

4.3.2 Framework Evaluation Criteria

Each framework was assessed across five operational dimensions:

- Functional Coverage — What categories of software behavior does it test?
- Data Processing Capability — Can it validate transformations and mathematical operations?
- Security Posture — Does it require external server access or expose internal data?
- Integration Compatibility — Does it work within BHEL's existing development environment?
- Maintenance Overhead — How much effort is required to write, update, and run tests?

4.3.3 Frameworks Evaluated

A. Selenium WebDriver

Selenium is an open-source browser automation framework used to validate user interface (UI) behavior in web-based applications. It simulates real user interactions — clicking buttons, filling forms, navigating menus — and captures runtime exceptions or unexpected outputs that indicate frontend logic errors.

At HERP, this is relevant for testing the web-based dashboards that production managers use to view KPI summaries. Selenium scripts can simulate a manager logging in, navigating to the inventory dashboard, and verifying that the displayed part count matches the expected database value — automating what would otherwise be a manual QA walkthrough before each deployment.

B. PyTest

PyTest is a Python-based unit and integration testing framework. It enables developers to define test functions that assert specific expected outcomes for each function in a script. For the data engineering pipelines developed at HERP, PyTest validates that transformation functions produce correct outputs — for example, confirming that a deduplication function correctly reduces a 10,200-row dataset with 200 duplicate entries to exactly 10,000 unique records.

PyTest also supports parameterized tests, allowing the same test logic to be run across dozens of input scenarios simultaneously — critical for catching edge cases in data cleaning functions that handle variable input quality from legacy sources.

C. Jenkins CI/CD Pipeline

Jenkins is an open-source continuous integration and continuous deployment (CI/CD) server. It automates the process of running test

suites whenever a developer commits new code changes to the repository. In BHEL's context, this means that any update to a production logging script automatically triggers PyTest and Selenium test suites, and the commit is only approved for deployment if all tests pass.

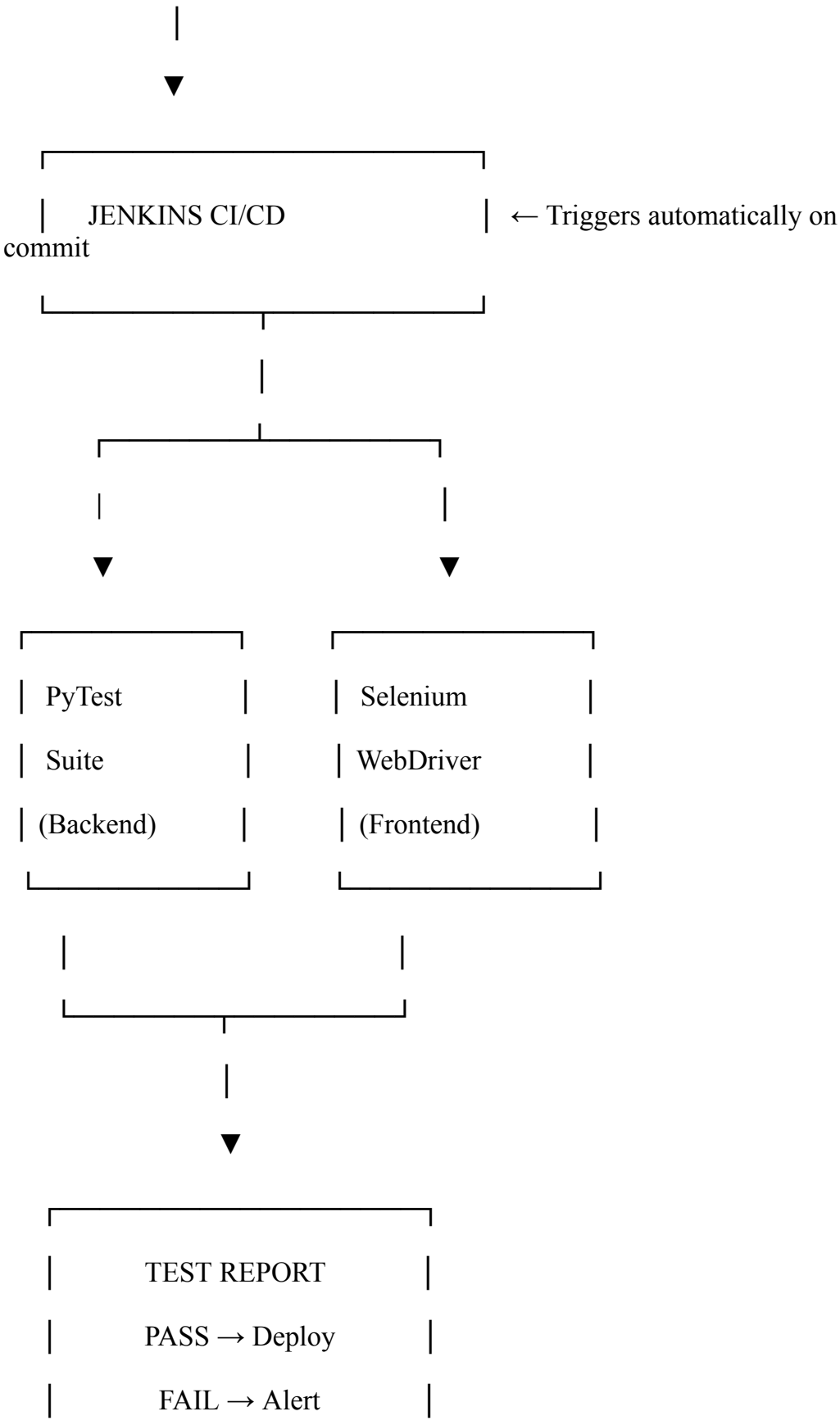
Jenkins is self-hosted — it runs on BHEL's internal servers without requiring connectivity to external cloud services. This is a critical requirement for an organization with strict data security policies around plant operational data.

4.3.5 Recommended Integration Architecture

Based on this evaluation, the recommended QA architecture for BHEL HERP's DX Group follows a layered testing model:

FIGURE 2: Layered QA Architecture

CODE COMMIT


JENKINS CI/CD
← Triggers automatically on commit
PyTest Suite (Backend)
Selenium WebDriver (Frontend)
TEST REPORT
PASS → Deploy
FAIL → Alert



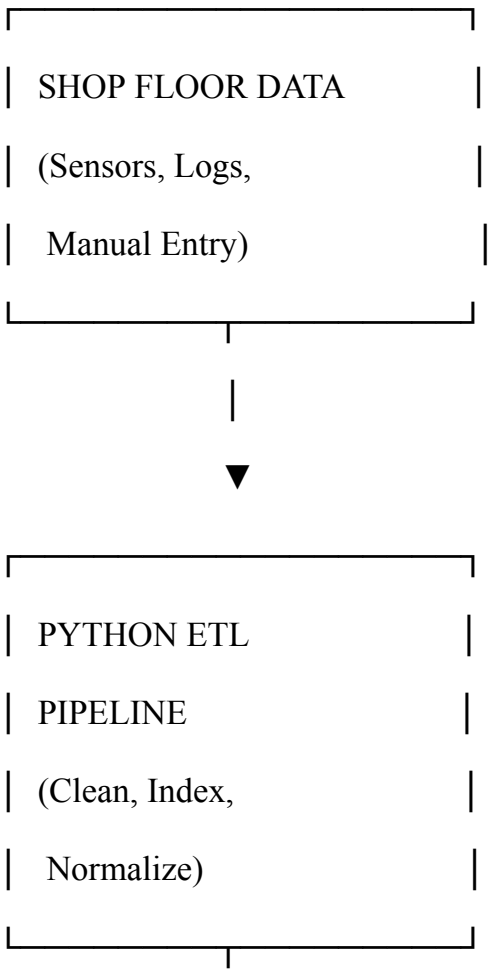
4.4 Automated Decision Support Architecture

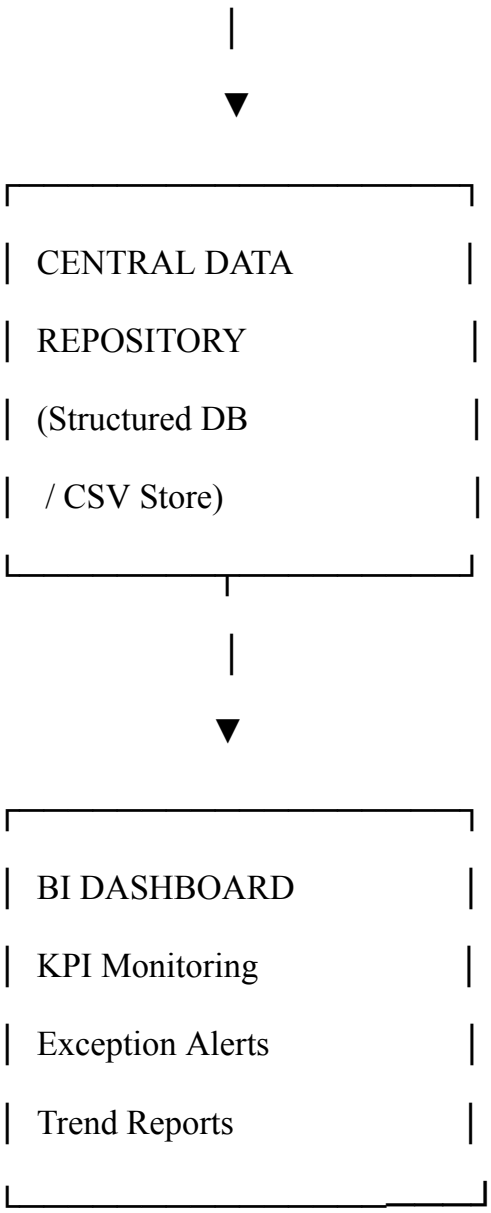
4.4.1 From Data Pipelines to Actionable Intelligence

Raw production logs — even when cleaned, indexed, and validated — are of limited operational value if they remain locked in database tables that only technical staff can access. The final phase of BHEL HERP's digital transformation initiative involves piping normalized operational data into Business Intelligence (BI) dashboards that surface key performance indicators (KPIs) to plant management in real time.

The decision support architecture designed during this internship follows a four-stage pipeline:

FIGURE 3: Decision Support Architecture — End-to-End Flow




CENTRAL DATA
REPOSITORY
(Structured DB
/ CSV Store)
BI DASHBOARD
KPI Monitoring
Exception Alerts
Trend Reports

5.RESULTS AND IMPACT ANALYSIS

5.1 Training Outcomes and Systems Competencies

The six-week industrial training tenure within the Digital Transformation Group at BHEL HERP Varanasi provided substantive, hands-on exposure to the application of modern computing frameworks within a large-scale industrial environment. Unlike academic coursework — where problems are well-bounded and datasets are clean — the work at HERP required navigating real-world constraints: legacy data quality issues, infrastructure limitations, strict security policies, and the need for solutions to be maintainable by a non-specialist plant IT team.

The key technical competencies developed across this period are detailed below.

5.1.1 Data Engineering and Automation

The core technical focus of the internship was the design of Python-based data engineering pipelines to automate HERP's production reporting workflows. This involved:

- Writing modular ETL scripts using Python and Pandas to extract, clean, and restructure fragmented spreadsheet data
- Implementing hash-indexed data structures to reduce query latency from O(n) to O(1)
- Developing parameterized data cleaning functions to handle inconsistent date formats, duplicate records, and null values systematically
- Documenting all script logic with inline comments and function docstrings to support future maintenance by the plant's internal team

The shift from manual spreadsheet reporting to automated pipelines

eliminated a category of human error that previously caused data discrepancies between shift records and management summaries.

5.1.2 Software Quality Assurance

A significant portion of the training focused on understanding how enterprise-grade software is validated before deployment. This included:

- Learning the structure and syntax of PyTest for writing unit tests and integration tests for data transformation functions
- Understanding how Selenium WebDriver automates UI validation workflows for web-based dashboards
- Studying Jenkins CI/CD pipeline configuration to appreciate how test automation integrates with code deployment cycles
- Evaluating the security posture of each framework relative to BHEL's internal data protection requirements

This QA exposure established a foundational understanding of why rigorous testing infrastructure is as important as the application code itself — particularly in environments where data errors carry real operational consequences.

5.1.3 Industrial Analytics and BI Systems

The final phase of the training involved contributing to the design of KPI monitoring dashboards:

- Identifying the operational KPIs most relevant to HERP's production management needs
- Mapping raw data fields from shop floor logs to dashboard metrics through transformation logic
- Understanding how exception-triggered alerts can proactively surface production anomalies before they escalate

- Learning how BI dashboards serve as the interface between raw operational data and management decision-making

5.1.4 Summary of Technical Skills Developed

TABLE 5: Technical Skills Acquired During Internship

Skill Area	Tools	Proficiency Developed
Data Engineering	Python, Pandas, NumPy	ETL pipeline design, data cleaning, indexing
Test Automation	PyTest, Selenium WebDriver	Unit testing, UI validation, test case design
CI/CD Integration	Jenkins	Automated pipeline configuration concepts
BI and Reporting	Dashboard design principles	KPI definition, exception alerting
Software Development Process	Agile SDLC	Sprint-based development, iterative delivery
Industrial Domain Knowledge	BHEL HERP operations	Manufacturing workflows, shop floor data systems

CONCLUSION

The industrial training programme at Bharat Heavy Electricals Limited (BHEL), Heavy Equipment Repair Plant, Varanasi, offered a rare and formative opportunity to observe, analyse, and contribute to the ongoing digital transformation of one of India's most significant public sector manufacturing institutions. The experience bridged a critical gap between academic study of computer science principles and their practical application under the architectural, security, and operational constraints of a heavy industry environment.

The primary technical contributions of this training — algorithmic optimization of production logs, systematic evaluation of open-source QA frameworks, and the design of an automated decision support pipeline — each addressed a specific operational inefficiency within HERP's existing digital infrastructure. The shift from linear O(n) spreadsheet lookups to O(1) indexed data structures demonstrated how foundational algorithmic principles have direct, measurable impact on enterprise process efficiency. The QA framework evaluation established a structured rationale for tool selection in environments where security and self-hostability are non-negotiable requirements. And the BI architecture work illustrated how cleaned, validated data only delivers value when it is surfaced in a form that supports real-time operational decision-making.

Beyond the technical domain, the training provided insight into how large public sector organizations balance the imperative to modernize against the constraints of legacy infrastructure, bureaucratic procurement cycles, and the need for solutions that can be maintained by plant teams rather than external vendors. The Agile methodology adopted by the DX Group offered one model for navigating this tension — delivering working improvements iteratively rather than waiting for comprehensive

system overhauls.

The software engineering practices, data pipeline design patterns, and industrial systems thinking developed during this programme establish a resilient foundation for professional responsibilities in technology-driven roles. The experience reinforced a core principle: that the value of technical knowledge is realized not in isolation, but through its disciplined application to problems that carry genuine consequence — and few environments make that consequence more tangible than the shop floor of a heavy manufacturing plant.

REFERENCES

- Listed by Alphabetical order.
- All references should be cited in the document without miss
- Format

[1] L. Xiao, C. Xie, M. Min and W. Zhuang, "User-Centric View of Unmanned Aerial Vehicle Transmission Against Smart Attacks", *IEEE Transactions on Vehicular Technology*, vol. 67, no. 4, pp. 3420-3430, April 2018.

APPENDIX I